

VR Project

Project Description

In this game, the player finds themselves in an abandoned chemistry lab. Around the room are various potions. The door is boarded up and various enemies are trying to get in. It's up to the player to launch potions at the enemy to defend themselves. The player can mix potions to create new ones that have different properties when thrown at an enemy.

Project Link

<https://github.com/Jasmine-Thind/vr-project>

User Interface

Press A to teleport to the next hotspot, and press the hand trigger while touching a potion to grab it. Throw and release the trigger to launch it.

One Line Description

Potion mixing and fighting game

External Assets

- The code handling the potion within its flask is taken from MinionsArt:
<https://www.patreon.com/posts/18245226>
- Chemistry Lab furniture from Unity Asset Store

Scripts

CountdownTimer.cs

EnemyAttack.cs

EnemyBargesIn.cs

EnemyChase.cs

EnemyHealth.cs

GameOver.cs

Liquid.cs

NextHotSpot.cs

PlayerHealth.cs

PotionCollision.cs

PourPotion.cs

PotionLiquid.shader

Descriptions

NextHotSpot.cs

- Enables hotspot teleportation
- Taken from lab 3

Liquid.cs

- Handles the fluid motion within the potion flasks
- Taken online from MinionsArt
- Comes with the shader PotionLiquid.shader from the same source

PourPotion.cs

- Creates the particles representing the pouring effect
- Based on the tilt angle (which varies based on the bottle shape), determine whether to start the pour flow
- Use a ParticleSystem to create the particles
- Lerp the potion fill amount to show it draining

PotionCollision.cs

- Triggered when the contents of one potion hit another
- Lerp the colour and show that the flask being poured into is filling

PotionEffect.cs

- Handle the logic of potions colliding with enemies
- Trigger different enemy functions based on the colour of the potion
- When the bottle collides, invoke a large stream of particles to act as the explosion of the potion

CountdownTimer.cs

- Displays a countdown timer in the player's view indicating time until the next enemy attack

- Timer resets after each enemy enters and disappears after the last enemy has entered

EnemyAttack.cs

- Handles damage dealt to the player when an enemy is in contact
- Reduces player health at a set rate while the player is within the enemy's trigger collider

EnemyBargesIn.cs

- Controls the full enemy entrance sequence after the timer expires
- Opens the door, moves the enemy to a designated endpoint, then closes the door
- Enables NavMesh chasing once the enemy reaches the endpoint
- Plays a door knock sound effect before the door opens

EnemyChase.cs

- Uses Unity's NavMesh AI to navigate the enemy toward the player's position
- Activated by EnemyBargesIn once the enemy has entered the room

EnemyHealth.cs

- Tracks enemy state and handles the three potion effects
- Slows the enemy's movement speed for a duration when hit by the red potion
- Disables NavMesh rotation and spins the enemy when hit by the green potion
- Destroys the enemy immediately when hit by the mixed potion

GameOver.cs

- Displays a game over screen when the player's health reaches zero
- Freezes the game and automatically restarts the scene after a short delay

PlayerHealth.cs

- Tracks the player's health and reduces it when taking damage from enemies
- Updates a red screen overlay that becomes more intense as health decreases
- Triggers the GameOver script when health reaches zero